

CATS⁺

Competishun Advance

Training Sessions

(Lecture - I)

Que

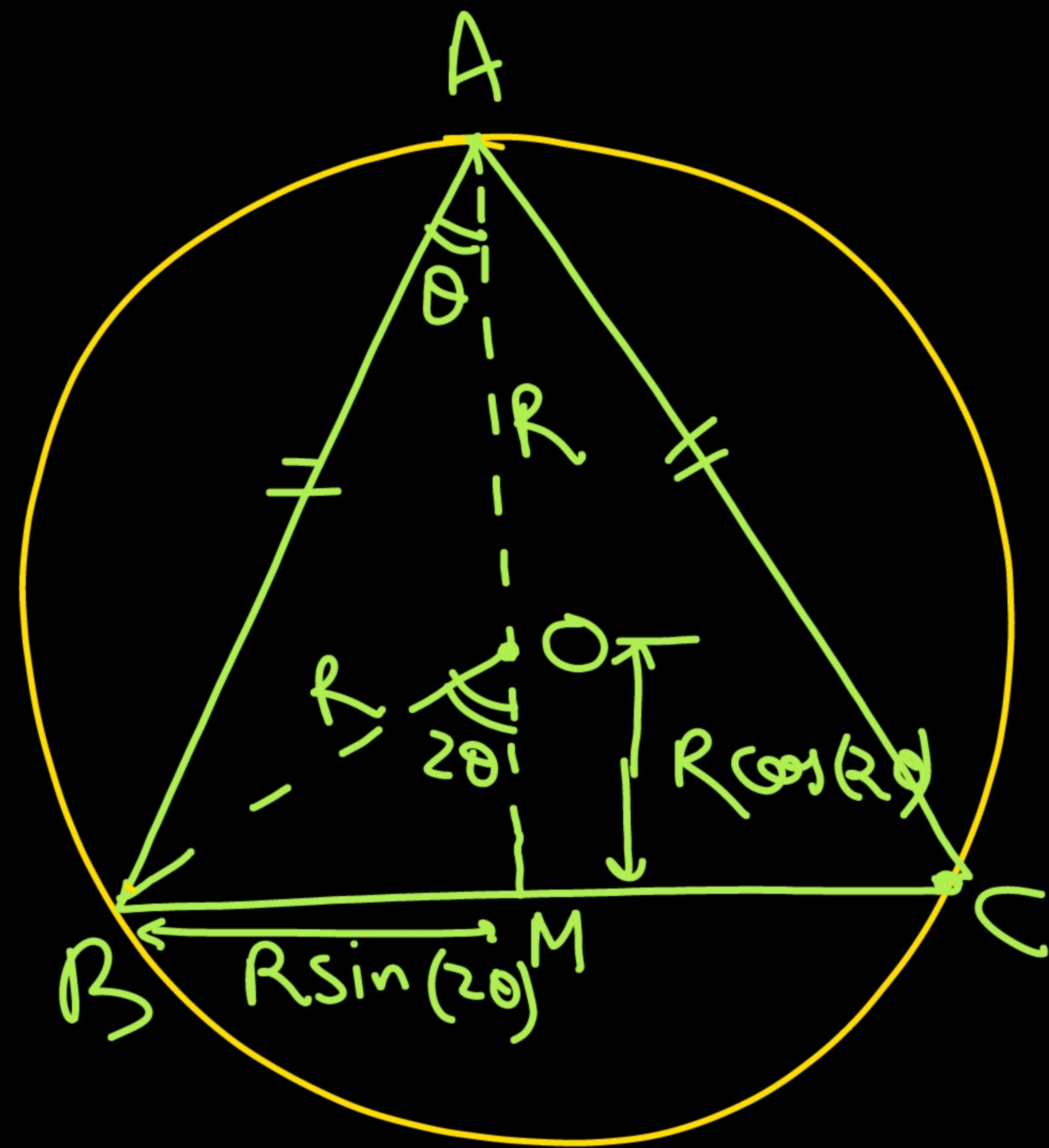
Let a circular disk has radius R , if it can cover a given isosceles triangle of perimeter L then value of R could be

☒ (A) $\frac{L}{2}$

☒ (B) $\frac{L}{3}$

☒ (C) $\frac{L}{4}$

(D) $\frac{L}{6}$



$$\theta \in \left(0, \frac{\pi}{2}\right)$$

$$\begin{aligned} L &= AB + BC + AC = 2(AB + BM) \\ &= 2 \left(R \sin(2\theta) \cdot \csc(\theta) + R \sin(2\theta) \right) \\ &= 2 \cdot R \sin(2\theta) \cdot (\csc(\theta) + 1) \\ &= 2 \cdot R \cancel{2 \sin(\theta)} \csc(\theta) \left(\frac{1 + \sin(\theta)}{\cancel{\sin(\theta)}} \right) \end{aligned}$$

$$L = 4R \cos(\theta) \cdot (1 + \sin(\theta))$$

$$R = \frac{L}{4 \cos(\theta) (1 + \sin(\theta))}$$

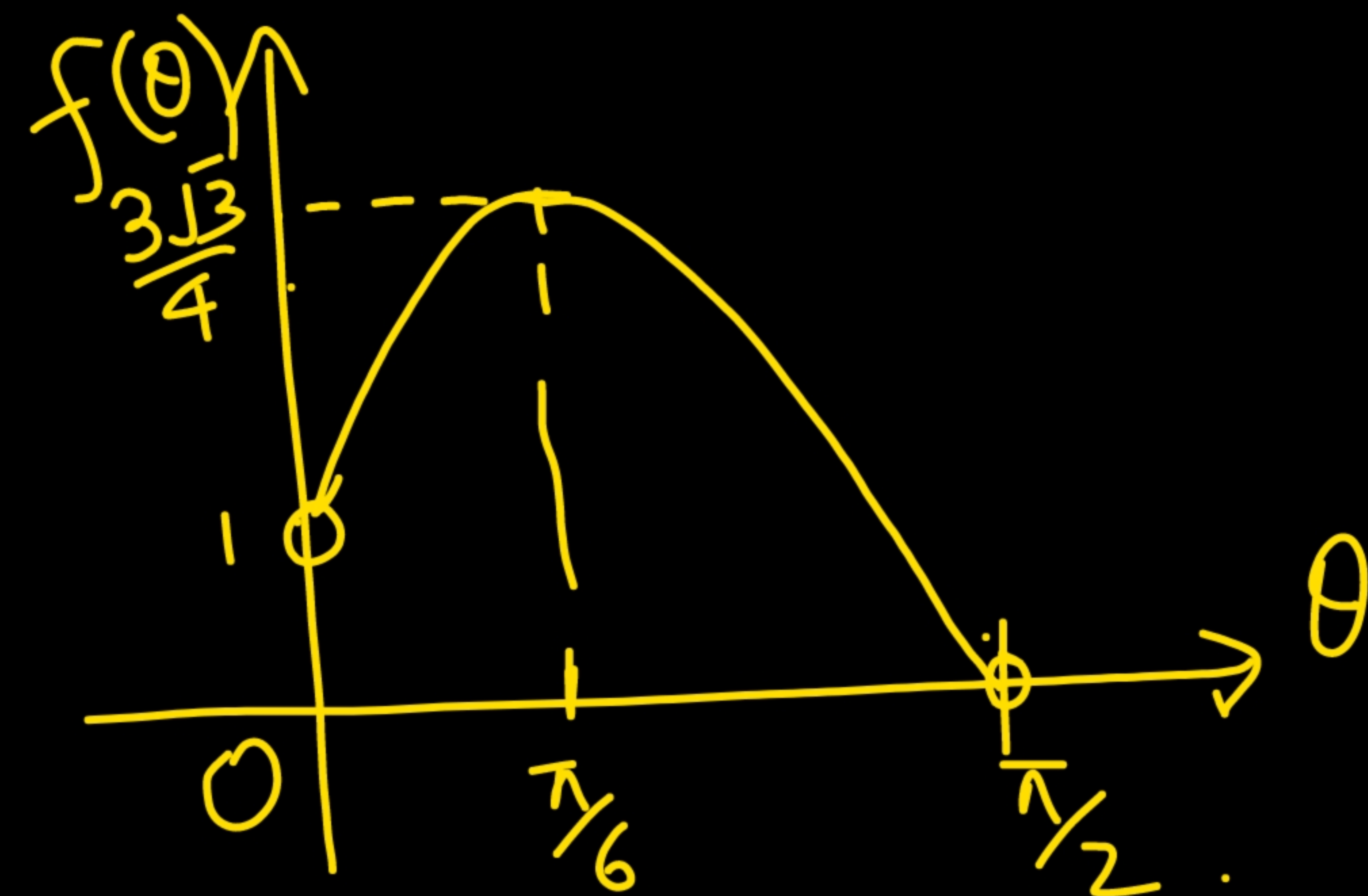
$$\text{let } f(\theta) = \cos(\theta) (1 + \sin(\theta))$$

$$f(\theta) = \cos \theta (1 + \sin \theta), \quad \theta \in (0, \frac{\pi}{2})$$

$$\begin{aligned} f'(\theta) &= -\sin \theta (1 + \sin \theta) + \cos \theta \cdot \cos \theta \\ &= -\sin \theta (1 + \sin \theta) + (1 - \sin^2 \theta) \\ &= (1 + \sin \theta)(1 - 2 \sin \theta) \\ &= 0 \end{aligned}$$

$$\theta = \frac{\pi}{6}$$

sign of $f'(\theta)$



$$f(\theta)_{\max} = \frac{\sqrt{3}}{2} \cdot \frac{3}{2} = \frac{3\sqrt{3}}{4}$$

$$R_{f(\theta)} = \left(0, \frac{3\sqrt{3}}{4}\right]$$

$$R_{\frac{1}{f(\theta)}} \in \left[\frac{4}{3\sqrt{3}}, \infty\right)$$

$$R = \frac{L}{4f(0)}$$

$$\in \left[\frac{4}{3\sqrt{3}} \cdot \frac{L}{4}, \infty \right)$$

$$\equiv \left[\frac{L}{3\sqrt{3}}, \infty \right)$$

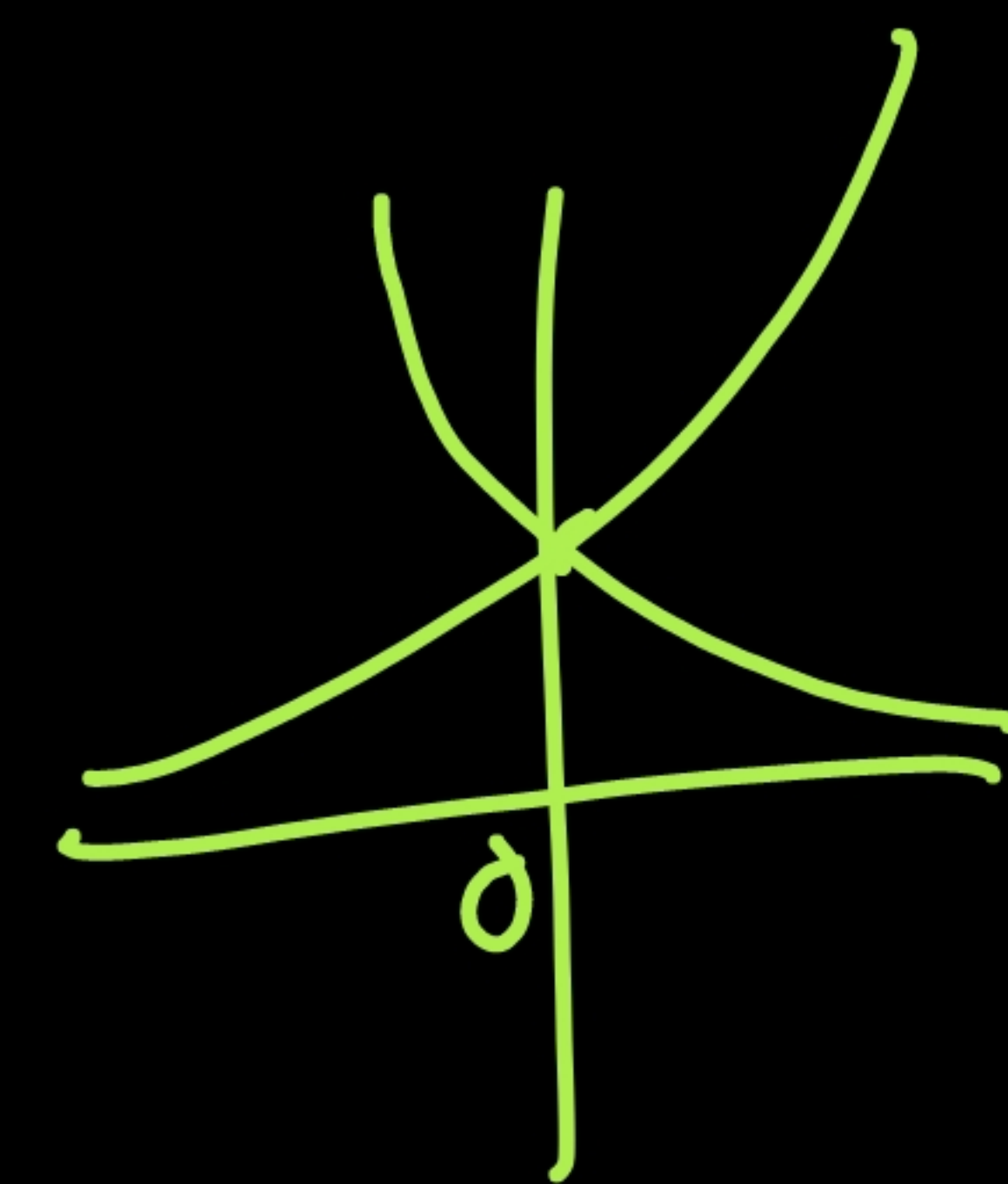
$$L \sim \frac{L}{5.1}$$

Que If $f(x)$ and $g(x)$ are two differentiable functions satisfying the following conditions (i) $f(0) = 2$ (ii) $g(0) = 1$ (iii) $f'(x) = g(x)$ and (iv) $g'(x) = f(x)$,
Now answer the followings.

Q Number of solutions of $f(x) = g(x)$ is
☒ A 0 ☐ B 1 ☐ C 2 ☐ D Infinite.

$$\begin{aligned} \frac{3}{2}e^x + \frac{1}{2}e^{-x} &= \frac{3}{2}e^x - \frac{1}{2}e^{-x} \\ e^{-x} &= 0 \quad \times \end{aligned}$$

Q Number of solutions of $f(x) = 2g(x)$ is
☐ A 0 ☒ B 1 ☐ C 2 ☐ D Infinite.



$$\begin{aligned} \frac{3}{2}e^x + \frac{1}{2}e^{-x} &= 3e^x - e^{-x} \\ e^x &= \frac{3}{2}e^{-x} \\ e^x &= e^{-x} \\ \Rightarrow x &= 0. \end{aligned}$$

$$\left[\begin{array}{l} f'(x) = g(x) \\ g'(x) = f(x) \end{array} \right] \xrightarrow{\cdot}$$

$$f'(x) - g'(x) = g(x) - f(x)$$

$$\Rightarrow \frac{d}{dx}(f - g) = -(f - g)$$

$$\left[\begin{array}{l} \frac{d(f-g)}{f-g} = -dx \\ \int \end{array} \right. \Rightarrow f(x) - g(x) = k_2 e^{-x}$$

$$\rightarrow f(x) - g(x) = k_2 e^{-x}$$

$$f(x) = \frac{k_1 e^x + k_2 e^{-x}}{2}$$

$$\rightarrow g(x) = \frac{k_1 e^x - k_2 e^{-x}}{2}$$

$$\Rightarrow f'(x) + g'(x) = g(x) + f(x)$$

$$\Rightarrow \frac{d}{dx}(f(x) + g(x)) = f(x) + g(x)$$

$$\left[\begin{array}{l} \frac{d(f+g)}{f+g} = dx \\ \int \end{array} \right. \Rightarrow \ln(f+g) = x + c$$

$$\Rightarrow \ln(f+g) = x + c$$

$$f(x) + g(x) = e^{x+c} = e^x \cdot e^c = k_1 e^x$$

$$\Rightarrow f(x) + g(x) = k_1 e^x$$

$$f(x) = \frac{k_1 e^x + k_2 e^{-x}}{2} \xrightarrow{x=0} \begin{cases} f(0) = \frac{k_1 + k_2}{2} \\ = 2 \end{cases} \Rightarrow k_1 + k_2 = 4$$

$$g(x) = \frac{k_1 e^x - k_2 e^{-x}}{2} \xrightarrow{x=0} \begin{cases} g(0) = \frac{k_1 - k_2}{2} \\ = 1 \end{cases} \Rightarrow \begin{aligned} &k_1 - k_2 = 2 \\ &+ \quad k_1 + k_2 = 4 \\ &\hline &k_1 = 3 \\ &k_2 = 1 \end{aligned}$$

$$f(x) = \frac{3}{2} e^x + \frac{1}{2} e^{-x} \quad \textcircled{8} \quad g(x) = \frac{3}{2} e^x - \frac{1}{2} e^{-x}$$

Ques

If $\cos 5\alpha = \cos^5 \alpha$, where $\alpha \in (0, \frac{\pi}{2})$, then sum of all the possible values of $(\sec^2 \alpha + \operatorname{cosec}^2 \alpha + \cot^2 \alpha)$ is equal to

- (A) 3 (B) 5 (C) 6 (D) None of these

$$\begin{aligned} \cos 5\alpha &= \cos^5 \alpha \\ \Rightarrow \cos(3\alpha + 2\alpha) &= \cos^5 \alpha \\ &= \cos 3\alpha \cdot \cos 2\alpha - \sin 3\alpha \cdot \sin 2\alpha \\ &= (4\cos^3 \alpha - 3\cos \alpha)(2\cos^2 \alpha - 1) - (3\sin \alpha - 4\sin^3 \alpha) 2\sin \alpha \cos \alpha \\ &= \cos \alpha (4\cos^2 \alpha - 3)(2\cos^2 \alpha - 1) - \sin^2 \alpha (3 - 4(1 - \cos^2 \alpha)) 2\cos \alpha \end{aligned}$$

$$\sec^2 \alpha + \operatorname{cosec}^2 \alpha + \cot^2 \alpha$$

$$\cos(5x) = \operatorname{Re} (e^{i5x}) = \frac{(e^{i5x})^5 + (e^{-i5x})^5}{2}$$

$$= \frac{(c + i s)^5 + (c - i s)^5}{2}$$

$$= {}^5C_0 \cdot c^5 - {}^5C_2 c^3 s^2 + {}^5C_4 c s^4$$

$$= c^5 - 10 c^3 s^2 + 5 c s^4$$

Given $\sin x = \frac{1}{3}$

$$\cos(5x) = \cos(x)$$

$$\Rightarrow \cancel{c^5} - \cancel{10} c^3 s^2 + \cancel{5} c s^4 = \cancel{c^5}$$

$$\Rightarrow 2 c^3 s^2 = \cancel{c s^4}$$

$$2c^2 = s^2 = 1 - c^2$$

$$3c^2 = 1 \Rightarrow \cos^2(x) = \frac{1}{3}$$

$$\sin^2(x) = \frac{2}{3}$$

$$\sec^2(x) + \csc^2(x) + \cot^2(x) = 3 + \frac{3}{2} + \frac{1}{2}$$

$$= \boxed{5}$$

One

Which of the following alternative(s) is/are correct?

✓ (A) $\tan^2 \frac{\pi}{7} + \tan^2 \frac{2\pi}{7} + \tan^2 \frac{3\pi}{7} = 21$

✓ (B) $\tan^2 \frac{\pi}{7} \tan^2 \frac{2\pi}{7} \tan^2 \frac{3\pi}{7} = 7$

(C) $\cot^2 \frac{\pi}{7} \cot^2 \frac{2\pi}{7} \cot^2 \frac{3\pi}{7} = \frac{1}{5}$ ✗

✓ (D) $\cot^2 \left(\frac{\pi}{7} \right) + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} = 5$

$$\sum \frac{1}{t_i^2} = \frac{\sum t_1^2 t_2^2}{t_1^2 t_2^2 t_3^2} = \frac{35}{7} = 5$$

Soln. Let $\frac{n\pi}{7} = \theta \Rightarrow n\pi = 7\theta = 4\theta + 3\theta \Rightarrow n\pi - 3\theta = 4\theta$
 $\tan(n\pi - 3\theta) = \tan(4\theta) \Rightarrow \tan(3\theta) = \tan(4\theta) \xrightarrow[\text{say}]{\tan(\theta)=t} -\frac{3t-t^3}{1-3t^2} = \frac{2t \tan(2\theta)}{1-\tan^2(2\theta)}$

$$\begin{aligned} \frac{t^3-3t}{1-3t^2} &= \frac{2\left(\frac{2t}{1-t^2}\right) - \left(\frac{4t}{1-t^2}\right)}{1 - \left(\frac{2t}{1-t^2}\right)^2} = \frac{4t(1-t^2)}{1+t^4-6t^2} \\ \Rightarrow \frac{t^2-3}{1-3t^2} &= \frac{4(1-t^2)}{t^4-6t^2+1} \Rightarrow (t^2-3)(t^4-6t^2+1) = 4(t^2-1)(3t^2-1) \end{aligned}$$

$$t^6 - 9t^4 + 19t^2 - 3 = 4(3t^4 - 4t^4 + 1)$$

$$z^3 - 21z^2 + 35z - 7 = 0$$

$$\hookrightarrow \tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right), \tan^2\left(\frac{3\pi}{7}\right)$$

$$\sum \tan^2\left(\frac{\pi}{7}\right) = 21$$

$$\sum \tan^2\left(\frac{\pi}{7}\right) \tan^2\left(\frac{2\pi}{7}\right) = 35$$

$$\prod \tan^2\left(\frac{\pi}{7}\right) = 7$$

Ques Which of the following alternative(s) is/are incorrect?

T (A) let $0 < \alpha, \beta, \gamma < \frac{\pi}{2}$. If $\cos \alpha = \alpha$, $\cos(\sin \beta) = \beta$, $\sin(\cos \gamma) = \gamma$ then $\gamma < \alpha < \beta$

F (B) Range of the function $f(x) = e^{\ln 3 \sin x} + e^{\ln 4 \cos x}$ is $[3, 5]$

T (C) $\cot(\sin \theta) > \tan(\cos \theta) \forall \theta \in \left(0, \frac{\pi}{2}\right)$

F (D) The value of $4\cos \frac{\pi}{10} - 3\sec \frac{\pi}{10} - 2\tan \frac{\pi}{10}$ is equal to 1

(A) $0 < \alpha, \beta, \gamma < \frac{\pi}{2}$.

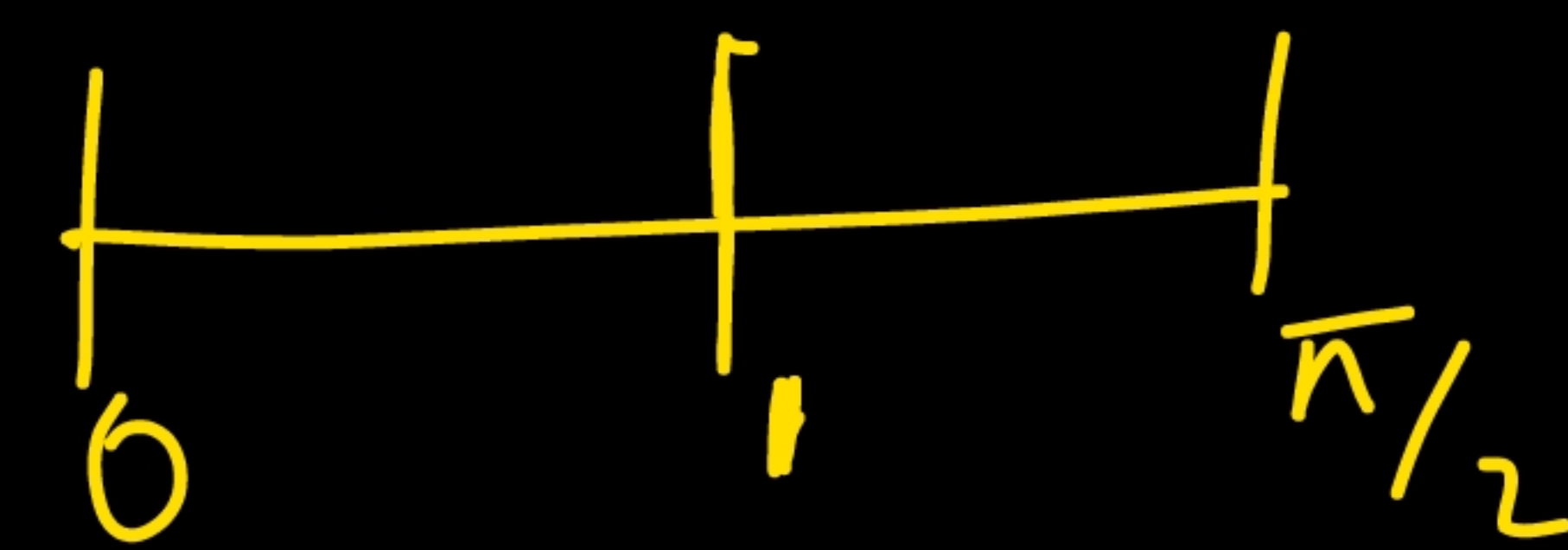
$$\cos(\alpha) = \alpha$$

$$\cos(\sin(\beta)) = \beta \Rightarrow \beta = \cos(\sin(\beta)) > \cos(\beta)$$

$$\sin(\cos(\gamma)) = \gamma \Rightarrow \gamma = \sin(\cos(\gamma)) < \cos(\gamma)$$

$$\gamma < \cos(\gamma)$$

Ans BD



$\sin(0, \pi/2)$

$$\sin(\theta) < \theta < \tan(\theta)$$

$$\sin(\beta) < \beta$$

As $\ln(0, \frac{\pi}{2})$
 $\cos(x)$ is $\downarrow$ ing fr. $\downarrow$

$$\cos(\sin(\beta)) > \cos(\beta)$$

Using $\sin(\cos(\gamma)) < \cos(\gamma)$

$$4 \cos\left(\frac{\pi}{10}\right) - 3 \sec\left(\frac{\pi}{10}\right) - 2 \tan\left(\frac{\pi}{10}\right)$$

$$= 4 \cos(\theta) - \frac{3}{\cos(\theta)} - \frac{2 \sin(\theta)}{\cos(\theta)}$$

$$= \frac{4 \cos^2(\theta) - 3 - 2 \sin(\theta)}{\cos(\theta)}$$

$$= \frac{4 \cos^3(\theta) - 3 \cos(\theta) - 2 \sin(\theta) \cos(\theta)}{\cos^2(\theta)}$$

$$= \frac{\cos\left(\frac{3\pi}{10}\right) - \sin\left(\frac{2\pi}{10}\right)}{\cos^2\left(\frac{\pi}{10}\right)} = \frac{\sin\left(\frac{2\pi}{10}\right) - \sin\left(\frac{2\pi}{10}\right)}{\cos^2\left(\frac{\pi}{10}\right)} = 0.$$

$$\cos(3\theta) = 4 \cos^3(\theta) - 3 \cos(\theta)$$